# Predicting Ames House Prices with a Bayesian Linear Model

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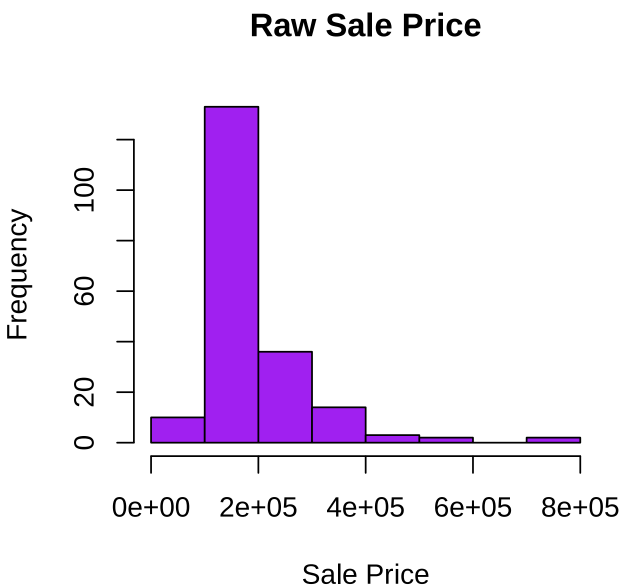
## Part (a):

### Exploratory Data Analysis (EDA):

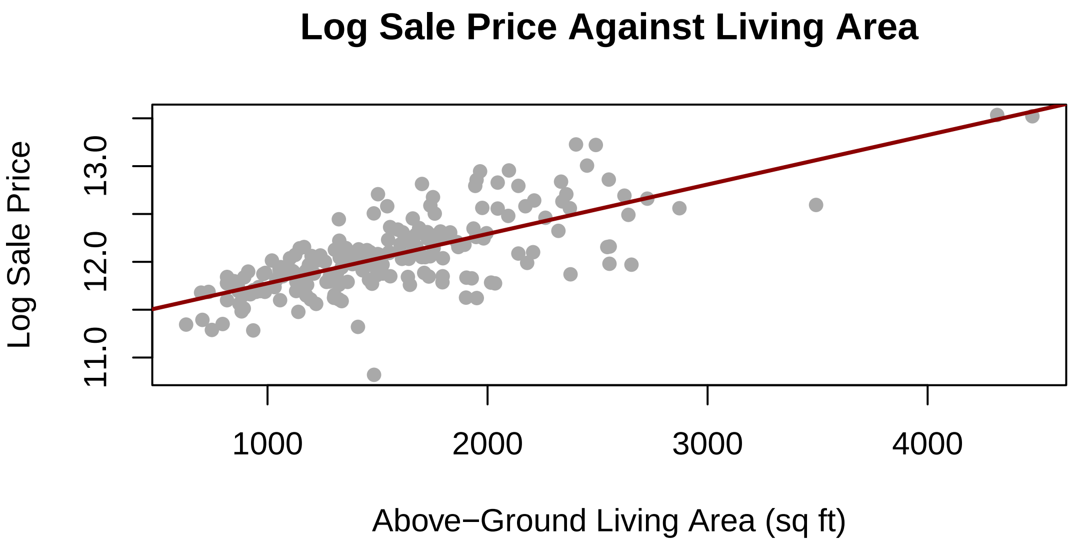
Before starting the exploratory data analysis and fitting any models,the data required cleaning up and processing. Specifically, Overall_Qual and Overall_Cond were converted into ordered factors, this is because it properly preserve their 1-10 ordinal ranking. Therefore, we can effectively analyse them when it comes to the EDA.

In this EDA, a scatter graph and two histograms will be used to gain a general picture of the data. These plots will be crucial for justifying our data transformations and for selecting what variable is the most effective to include in the proposed Bayesian Normal linear models.The histograms will visualise the distribution of the response variable and justify any need of transformation. The scatter graph will give an insight into the linear trend between the house price and our strongest contentious predictor.





As illustrated in the histograms above, the histogram that is showing the raw sale price (left) has a clear and prominent positive right skew. To correct this skew a log transformation was used to descend the ladder of powers. The second histogram portrays a log-transformed interpretation of the data, which is far more symmetric and normally distributed.




In this scatter graph, the newly transformed log_Sale_Price is plotted against the above-ground living area (Gr_Liv_Area). A clear positive linear relationship is displayed with the data points cluster closely around the line of best fit. This indicates that as the living area increases, the log sale price increases at a steadily linear rate. This confirms that living area is a highly influential predictor, therefore it is a vital variable to include in the Bayesian Normal linear models.

### Model Selection:

To identify the most effective Bayesian Normal linear model for predicting the transformed house price, three candidate models have been proposed where is more complex than the previous. As analysed in the EDA, the above-ground living area (Gr_Liv_Area) variable is included in all three models as the baselines continuous predictor due to its obvious correlation with house prices. To test if the predictive power improves, more prepossessed ordinal variables will be added to each model.

The models are as follows:

- Model 1: log_Sale_Price ~ Gr_Liv_Area (The baseline model just using the strongest continuous predictor. This is the most simple model).
- Model 2: log_Sale_Price ~ Gr_Liv_Area + Overall_Qual (Adding the overall material and finish quality to the baselines model).
- Model 3: log_Sale_Price ~ Gr_Liv_Area + Overall_Qual + Overall_Cond (Adding overall condition rating to the second model. This is the most complex model).

```
## [1] "BIC Values:"
```

```
##       df       BIC
## m1_lm  3  74.85812
## m2_lm 10 -63.76537
## m3_lm 16 -59.42823
```

```
## [1] "Bayes Factor (Model 2 vs Model 1)"
```

```
## Bayes factor analysis
## --------------
## [1] Gr_Liv_Area + Overall_Qual : 1.427232e+29 ±0.71%
##
## Against denominator:
##   log_Sale_Price ~ Gr_Liv_Area
## ---
```

```
## Bayes factor type: BFlinearModel, JZS

## [1] "Bayes Factor (Model 3 vs Model 2)"

## Bayes factor analysis
## --------------
## [1] Gr_Liv_Area + Overall_Qual + Overall_Cond : 1.079346 ±2.21%
##
## Against denominator:
##   log_Sale_Price ~ Gr_Liv_Area + Overall_Qual
## ---
## Bayes factor type: BFlinearModel, JZS
```

To choose the most optimal model, the Bayesian Information Criterion (BIC) was generated. This acts as a model selection criterion, balancing goodness-of-fit while also noting model complexity to prevent overfitting. From the analysis performed by the BayesFactor package, we can see that Model 2 outperforms both the simpler model (Model 1) and the more complex model (Model 3). This is due to it yielding the lowest BIC (-63.76537) in comparison to Model 1 (74.85812) and Model 3 (-59.42823).

These findings are also strongly supported by the Bayes Factor analysis. This quantifies the relative evidence of one model over another. The analysis computed above finds the Bayes Factor for Model 2 Vs Model 1 to be approximately 1.420716e+29±0.85% (roughly 1.42*10^29). According to the Bayes Factor analysis, any Bayes Factor>10 provides evidence in favor of the more complex model, this therefore confirms that the Overall_Qual variable is a vital predictor. Looking at the Bayes Factor for Model 3 Vs Model 2, the Bayes Factor is approximately 1.075202±1.92%. Since this value is between 1 and 3 it provides weak evidence that Overall_Cond is not a very effective predictor and adding this complexity is unjustified. Therefore according the the Bayes Factor, Model 2 is the most effective as it adds just enough complexity without the risk of overfitting.

In conclusion, both the BIC and the Bayes Factor analysis point towards using Model 2 (log_Sale_Price ~ Gr_Liv_Area + Overall_Qual) being the most effective optimal choice. This is due to its low BIC and its optimal Bayes Factor analysis score when compared to other models.

### Selected Model Posterior

For the Bayesian Normal linear model chosen (Model 2), the joint posterior distribution can be expressed in terms of the regressions coefficients ($\beta$) and the error variance ($\sigma^2$). The posterior distributions, based on the conjugate Normal-Inverse-Gamma prior, are:

$$\beta|\mathbf{y},\mathbf{Z},\sigma^2 \sim N(\mathbf{b}_n, \sigma^2\mathbf{\Sigma}_n)$$

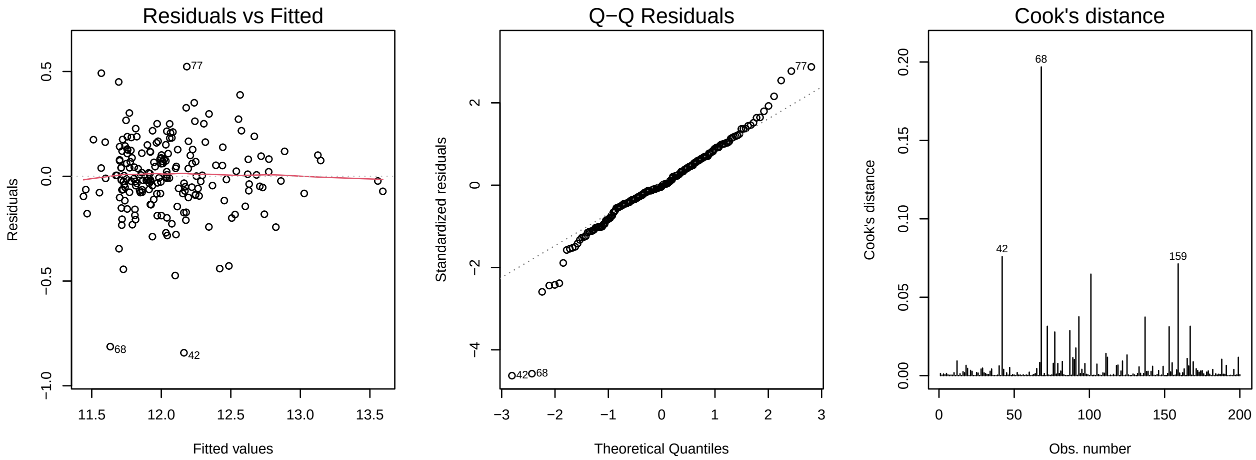
$$\sigma^2|\mathbf{y},\mathbf{Z} \sim IG(\mathbf{a}_n, \mathbf{b}_n)$$

Here $\mathbf{b}_n$ and $\mathbf{\Sigma}_n$ are the updated posterior mean vector and the covariance matrix for all the coefficients. $a_n$ and $b_n$ both represent the updated shape and scale parameters for the Inverse-Gamma distribution of the variance. Specifically, these updated parameters are now defined as:

$$\mathbf{\Sigma}_n = (\mathbf{Z}^T\mathbf{Z} + \mathbf{\Sigma}_0^{-1})^{-1}$$
$$\mathbf{b}_n = \mathbf{\Sigma}_n(\mathbf{Z}^T\mathbf{y} + \mathbf{\Sigma}_0^{-1}\mathbf{b}_0)$$

$$a_n = a_0 + \frac{n}{2}$$

$$b_n = b_0 + \frac{1}{2}(\mathbf{y}^T\mathbf{y} + \mathbf{b}_0^T\mathbf{\Sigma}_0^{-1}\mathbf{b}_0 - \mathbf{b}_n^T\mathbf{\Sigma}_n^{-1}\mathbf{b}_n)$$




**Interpretations of Graphs**

1. Residuals vs Fitted: This diagram check for constant variance. This plot asses the assumption of homoscedasticity (the error terms should have a constant variance with no obvious trend). In the above plot, the red line is relatively flat near the horizontal 0.0 line. This indicates that the variance is generally constant. In the plot it is also important to note that although the data points are more clustered to the left of the diagram, this simply indicates the natural distribution of the housing data. Overall, the results from the Residual vs Fitted graph confirms the decision to log-transform the response variable was successful in stabalising the variance.

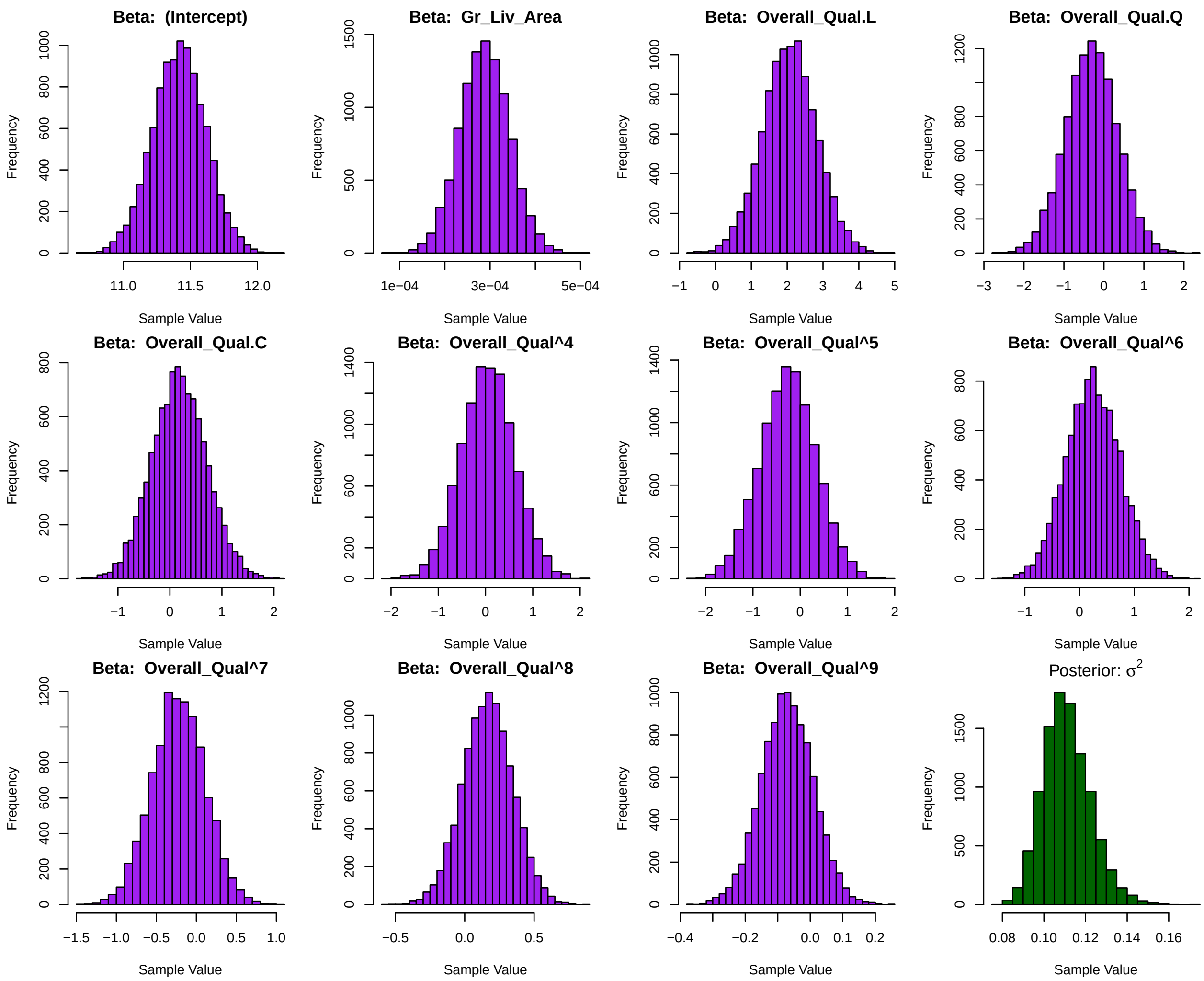
2. Q-Q Residuals: This diagram checks for weather or not the residuals from the model follow a normal distribution. From the second diagram above, it is clear that most of the data points follow a strong positive linear relationship due to most of said points following the theoretical line of best fit. The plot shows some points towards the tails of line of best fit deviating away, the assumption of normality holds sufficiently well for the majority of the data points.

3. Cook's distance: This diagram is used to identify any high-leverage outlines that may disproportionately distort the model parameters. The general rule of thumb to identify these outlines is to look for values with a Crook's distance greater than or equal to 1. In the above third plot, there are three values highlighted (42, 68 and 159). However, their values are nowhere near the threshold of 1.0, therefore it can be concluded that there are no high-leverage outliers that may disproportionately distort the model parameters.

# Part b:

## Posterior Sampling

To be able to conduct inference on the selected model (Model 2),samples will be drawn from its joint posterior distribution. In Part (a), the model parameters ($\beta$ and $\sigma^2$) follow a conjugate Normal-Inverse-Gamma distribution. Since these distributions are known, the updated posterior parameters ($\mathbf{b}_n$, $\mathbf{\Sigma}_n$, $a_n$, $b_n$) can be directly computed based on a vague prior and draw 10,000 samples using the MASS package. Firstly, the variance is sampled from the Inverse-Gamma distribution and then the coefficients are sampled from the Multivariate Normal distribution conditional on those $\sigma^2$ values.




### Interpretation of Posterior Distribution

The above grid of histograms visualises the marginal posterior distributions for the variance ($\sigma^2$) and each regression coefficients ($\beta$). As expected from the theoretical properties of the conjugate Normal-Inverse-Gamma prior, the sampled $\beta$ coefficient all exhibit symmetric, bell-shaped normal distributions around their respective means. Since Overall_Qual is an ordinal factor, R created a dummy variable for its different levels during the creation of the design matrix. This allows the district, isolated marginal distribution for each quality rating to be observed in comparison to the baseline.

Looking at the $\sigma^2$ histogram, a slight left-skew can be observed. However, it remains strictly positive and tightly centered, this reflects the strong updated certainty about the model's residual variance after accounting for the data.

## OLS Comparison

To evaluate the Bayesian model chosen, the point estimates (posterior means) of the sampled parameters will be computed and compares directly to the standard frequentist Ordinary Least Squares (OLS) estimates.The residual variance from the OLS model will be compared to the mean of the sampled Bayesian posterior variance.

```
## [1] "Comparion of Regression Coefficents:"
```

```
##                   OLSest Bayesest   Diff
## Z(Intercept)     12.9849  11.4175 1.5674
## ZGr_Liv_Area      0.0002   0.0003 0.0001
## ZOverall_Qual.L -4.6356   2.0671 6.7028
## ZOverall_Qual.Q  6.2156  -0.3071 6.5227
## ZOverall_Qual.C -4.4200   0.1824 4.6024
## ZOverall_Qual^4  2.5606   0.0491 2.5114
## ZOverall_Qual^5 -1.1186  -0.2581 0.8605
## ZOverall_Qual^6  0.2841   0.2457 0.0385
## ZOverall_Qual^7 -0.0006  -0.2239 0.2233
## ZOverall_Qual^8      NA   0.1650     NA
## ZOverall_Qual^9      NA  -0.0707     NA
```

```
## [1] "OLS Residual Variance: 0.0342"
```

```
## [1] "Bayesian Posterior Varience: 0.111"
```

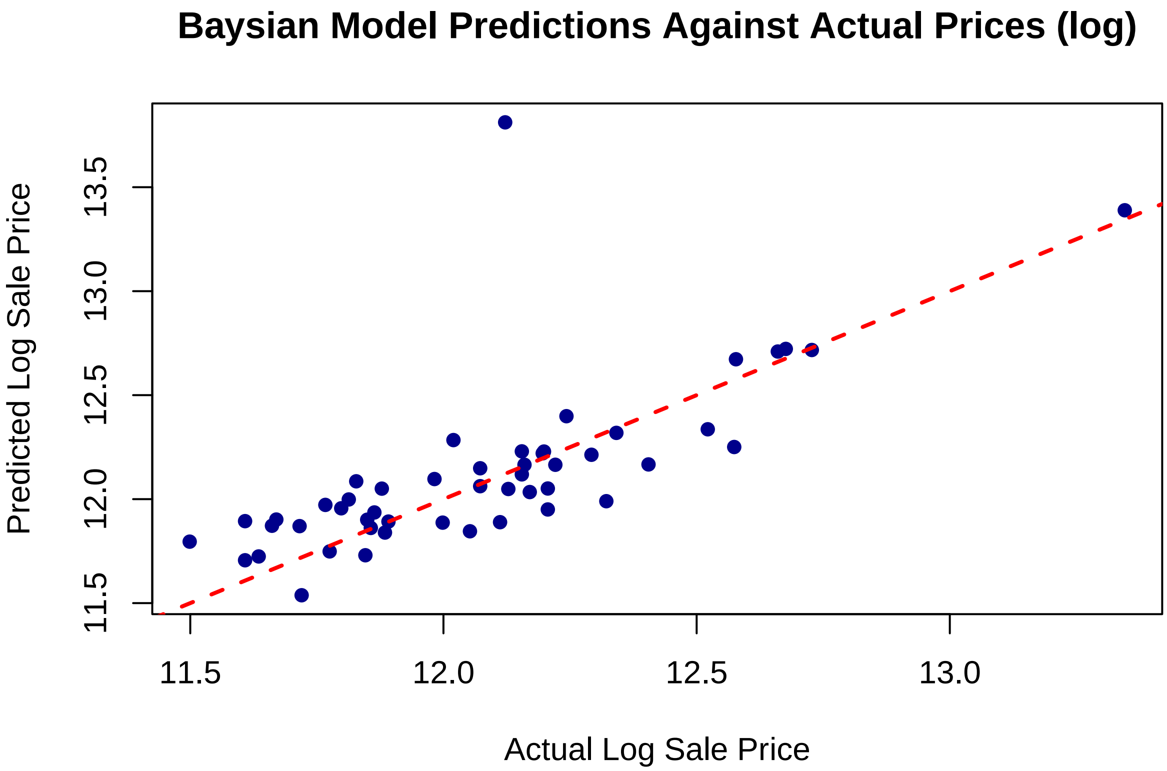
### Discussion of Results

As shown in the output table above, while the approximations for the baselines continuous predictor (Gr_Liv_Area) are more or less identical, there are some fairly district differences between the Bayesian point estimates and the OLS ones for the Overall_Qual dummy variables and intercept. The table also shows thar the OLS Residual Variance is smaller than the Bayesian Posterior Varience (0.0342 and 0.111 respectively).

Initially the assumed prior was vague (using a variance of 10 in the prior covariance matrix $\mathbf{\Sigma}_0$). The output now gained demonstrates that it is actually had a noticeable regularising effect. Since the prior was centered at mean 0, the true magnitude of the coefficients for the predictor Overall_Qual's levels is relatively large (ranging between -4.6 to 6.2 in the upenalised OLS model), this shows that the priot $\sigma$ (~3.16) was not large enough to be completely a truly vague prior. The consequence of this is that the Baysian model had applied a strong "shrinkage" effect that pulled the posterior estimates for the factor levels that are closer to 0. Furthermore, since the training dataset contains only 200 observations, the empirical data was not strong enough to overright this prior assumption. This leads onto the observed differences between the two sets of estimates.

## Test Set of Predictors

To be able to evaluate the predictive performance of our Bayesian model, the posterior mean estimates will be used to predict the house prices for the unseen data (AmesTest.csv). Firstly, the test data must be preprocessed identically to the training data by ordering the factor levels of Overall_Qual correctly. Then the Root Mean Squared Error (RMSE) is computed between the predicted log sale prices and true values and a scatter graph visualises the model's accuracy.




```
## [1] "Test Set RMSE:  0.2861"
```

**Conclusion of Predictive Performance**

The test set RMSE score was roughly 0.2861 which quantifies the average deviation of the Bayesian model's predictions from the true log sale prices. This score indicates a solid predictive accuracy for a parsimonious model relying only on two predictors (in this case Gr_Liv_Area and Overall_Qual). This is because, considering the response variable was log-transformed, an error of this level is fairly good for accuracy.

The diagram above confirms this strong performance. The blue data points follow the red line of perfect prediction fairley perfectly. This indicates that the model successfully generalises to new unseen data without too much overfitting. While a lot of the blue data points are clustered towards the left of the red line, this is merely reflects the natural distribution of housing data. The existence of the one district outlier at the top of the graph highlights a minor issue in the chosen model. This point represents a property that the model overpredicted because it only uses two predictors. However since most of the data points follow the red line, the Bayseian linear model serves as a highly effective tool for predicting standard Ames house prices without overfitting.

## Appendix

```r
library(ggplot2)
library(gridExtra)

ames_train <- read.csv("AmesTrain.csv")
ames_test <- read.csv("AmesTest.csv")

category_variables <- c("Overall_Qual", "Overall_Cond")
quality_levels <- c("Very_Poor", "Poor", "Fair", "Below_Average", "Average",
                    "Above_Average", "Good", "Very_Good", "Excellent", "Very_Excellent")

ames_train[category_variables] <- lapply(ames_train[category_variables], factor,
                                         levels = quality_levels, ordered =TRUE)
ames_test[category_variables] <- lapply(ames_test[category_variables], factor,
                                        levels = quality_levels, ordered =TRUE)


ames_train$log_Sale_Price <- log(ames_train$Sale_Price)
par(mfrow=c(1,2))
hist(ames_train$Sale_Price, main="Raw Sale Price", xlab="Sale Price", col="purple")
hist(ames_train$log_Sale_Price, main="Log Sale Price", xlab="Log Sale Price", col="red")
par(mfrow=c(1,1))


plot(ames_train$Gr_Liv_Area, ames_train$log_Sale_Price,
     main = "Log Sale Price Against Living Area", xlab="Above-Ground Living Area (sq ft)",
     ylab="Log Sale Price", pch=16, col="darkgray")

abline(lm(log_Sale_Price ~ Gr_Liv_Area, data =ames_train), col="darkred", lwd=2)

library(BayesFactor)

m1_lm <- lm(log_Sale_Price ~ Gr_Liv_Area, data = ames_train)
m2_lm <- lm(log_Sale_Price ~ Gr_Liv_Area + Overall_Qual, data = ames_train)
m3_lm <- lm(log_Sale_Price ~ Gr_Liv_Area + Overall_Qual + Overall_Cond, data = ames_train)


m1_bf <- lmBF(log_Sale_Price ~ Gr_Liv_Area, data = ames_train)
m2_bf <- lmBF(log_Sale_Price ~ Gr_Liv_Area + Overall_Qual, data= ames_train)
m3_bf <- lmBF(log_Sale_Price ~ Gr_Liv_Area + Overall_Qual+ Overall_Cond, data= ames_train)

print("BIC Values:")
BIC(m1_lm, m2_lm, m3_lm)

print("Bayes Factor (Model 2 vs Model 1)")
m2_bf/m1_bf

print("Bayes Factor (Model 3 vs Model 2)")
m3_bf/m2_bf

par(mfrow = c(1, 3))
plot(m2_lm, which = 1)
```

```r
plot(m2_lm, which = 2)
plot(m2_lm, which = 4)

library(MASS)

y<-ames_train$log_Sale_Price
Z<-model.matrix(~Gr_Liv_Area + Overall_Qual, data = ames_train)
n<-nrow(Z)
p<-ncol(Z)
B0 <- rep(0,p)
Sigma0 <- diag(10,p)
a0<- 1
b0<-1

Sigma0inv <-solve(Sigma0)
Sigman <- solve(Sigma0inv + t(Z) %*% Z)
Bn <- Sigman %*%(t(Z) %*% y +Sigma0inv %*% B0)

an <- a0+n/2
B0S0invB0 <- t(B0) %*% Sigma0inv %*% B0
BnSninvBn <- t(Bn) %*% solve(Sigman) %*% Bn
bn <- as.numeric(b0 +0.5*(t(y) %*% y + B0S0invB0 - BnSninvBn))

set.seed(42)
nsamples<-10000
sigma2samples <- 1/ rgamma(nsamples, shape=an, rate=bn)
betasamples <- matrix(0, nrow=nsamples, ncol=p)

for (i in 1:nsamples) { betasamples[i, ] <- mvrnorm(1, mu=Bn, Sigma = Sigman * sigma2samples[i])}

par(mfrow = c(3,4), mar = c(4,4,2,1))

for (j in 1:p) {
  hist(betasamples[, j],
       main = paste("Beta: ", colnames(Z)[j]),
       xlab = "Sample Value",
       col = "purple",
       breaks = 30)
}

hist(sigma2samples,
     main= expression(paste("Posterior: ", sigma^2)),
     xlab= "Sample Value",
     col="darkgreen",
     breaks=30)

bayescoef <- colMeans(betasamples)
bayessigma2 <- mean(sigma2samples)

olsmodel <- lm(y~Z-1)
olscoef <- coef(olsmodel)
olssigma2 <- summary(olsmodel)$sigma^2
```

```r
comparisondf <- data.frame(
  OLSest = round(as.numeric(olscoef), 4),
  Bayesest = round(as.numeric(bayescoef), 4),
  Diff = round(abs(as.numeric(olscoef) - as.numeric(bayescoef)), 4)
)

rownames(comparisondf) <- names(olscoef)

print("Comparion of Regression Coefficents:")
print(comparisondf)

print(paste("OLS Residual Variance:", round(olssigma2, 4)))
print(paste("Bayesian Posterior Varience:", round(bayessigma2, 4)))


ames_test$Overall_Qual <- factor(ames_test$Overall_Qual, levels=quality_levels, ordered=TRUE)
ytest <- log(ames_test$Sale_Price)

Ztest <- model.matrix(~Gr_Liv_Area + Overall_Qual, data = ames_test)

predictions <- Ztest %*% bayescoef

rmse <- sqrt(mean((ytest-predictions)^2))

plot(ytest, predictions,
     main = "Baysian Model Predictions Against Actual Prices (log)",
     xlab = "Actual Log Sale Price",
     ylab = "Predicted Log Sale Price",
     col = "darkblue", pch=16)

abline(0, 1, col = "red", lwd=2, lty = 2)

print(paste("Test Set RMSE: ", round(rmse, 4)))
```